BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 2 Packet.

Bone Tissue · Axial Skeleton · Appendicular Skeleton · Joints

BIO 004 · MODULE 2

Contents

Six chapters, in the order the module teaches them. Part A of the Module 2 packet. Part B is the pre-work, Part C is the lab structure list.

Everything else for Module 2

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 2 COVERS

Module 2 covers bone tissue, the axial and appendicular skeleton, and the joints that connect them. Muscle structure and the sarcomere belong to Module 3.

The six chapters build in order. Bone tissue is the material. The skull, spine, and thoracic cage are the axial frame built from it. The limbs and their girdles hang off that frame. Joints are where any two of these meet, so that chapter only makes sense once you can name the bones on both sides of the joint.

This is anatomy. Where function appears in these notes it is marked as a physiology note, and it is there because the structure does not make sense without it.

PART A

Course Notes.

Five chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B. This is your reference for every video, lab, and exam question in Module 1.

CHAPTER 1

Bone Tissue and Microanatomy

BONE HISTOLOGY

Bone and cartilage are the supportive connective tissues. This chapter covers cartilage, the structure of a bone from the whole organ down to the osteon, the four bone cells, and how bone forms and grows.

BY THE END

1. Describe the three types of cartilage and the cells and matrix that make up cartilage.
2. Classify bones by shape and identify the gross parts of a long bone.
3. Describe the microanatomy of compact bone, centred on the osteon, and contrast it with spongy bone.
4. Identify the four bone cell types and what each one does.
5. Outline how bones form and grow, using the growth plate and its zones in order.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Bone

drsrennie-stack.github.io/new-build-bio4-solano/bone-concept-videos.html

Cartilage

Cartilage is a firm, flexible connective tissue. It has no blood vessels of its own, so it heals slowly. That single fact explains most of what is clinically interesting about it.

THE CARTILAGE TISSUE

Component	What it is
Chondrocytes	The cartilage cells. Each one sits in a small space called a lacuna
Lacunae	The cavities in the matrix that house the chondrocytes
Matrix	A firm gel of ground substance with collagen, and in some types elastic fibers
Perichondrium	The connective tissue sheath wrapped around most cartilage

The three types of cartilage

FIBERS, TEXTURE, AND WHERE EACH IS FOUND

Type	Fibers and texture	Where it is found
Hyaline cartilage	Fine collagen, glassy and smooth. The most common type	Ends of long bones, the nose, the airways, the costal cartilages
Elastic cartilage	Many elastic fibers, flexible and springy	The external ear and the epiglottis
Fibrocartilage	Thick bundles of collagen, tough and shock-absorbing	The intervertebral discs, the knee menisci, the pubic symphysis

How cartilage grows

TWO DIRECTIONS OF GROWTH

Type of growth	What happens	Where it starts
Appositional	New cartilage is added at the outer surface	From the perichondrium
Interstitial	The cartilage expands from within	Chondrocytes divide inside the matrix

Bone Classification and the Long Bone

Bones by shape

FIVE SHAPES, WITH EXAMPLES

Shape	Description	Examples
Long	Longer than they are wide, a shaft with two ends	Femur, humerus
Short	Roughly cube-shaped	Carpals, tarsals
Flat	Thin and often curved	Sternum, ribs, most skull bones
Irregular	Complex shapes that fit no other group	Vertebrae, hip bones
Sesamoid	Small bones formed within a tendon	Patella

Gross anatomy of a long bone

FROM THE SHAFT OUTWARD

Structure	What it is
Diaphysis	The shaft, a tube of compact bone around the medullary cavity
Epiphysis	Each expanded end. Spongy bone under a thin compact shell
Metaphysis	The region between diaphysis and epiphysis. Holds the growth plate or its remnant
Medullary cavity	The hollow core of the diaphysis, holds marrow
Articular cartilage	Hyaline cartilage capping the epiphysis where it meets another bone
Periosteum	The connective tissue membrane covering the outer bone surface
Perforating (Sharpey) fibers	Bundles of collagen that anchor the periosteum into the underlying bone
Nutrient foramen	The opening in the shaft where the nutrient artery enters the bone
Endosteum	The thin membrane lining the internal bone surfaces
Epiphyseal line	The remnant of the growth plate after growth in length ends
Red marrow	Fills the spongy bone of the epiphyses. The site of blood cell formation
Yellow marrow	Fat stored in the medullary cavity of the adult shaft

Bone Microanatomy

Compact bone

THE OSTEON AND WHAT SURROUNDS IT

Structure	What it is
Osteon	The Haversian system, the structural unit of compact bone. A set of tubes around a canal
Central canal	The Haversian canal at the core of the osteon. Carries blood vessels and nerves
Perforating canal	The Volkmann canal. Runs crosswise and links central canals
Lamellae	The concentric rings of bony matrix around the central canal
Lacunae	Small cavities between the lamellae. Each holds one osteocyte
Canaliculi	Tiny channels that connect lacunae and let osteocytes exchange materials
Interstitial lamellae	Leftover fragments of old osteons, filling the gaps between intact osteons
Circumferential lamellae	Rings of bone that wrap the entire outer and inner surface of the shaft

Spongy bone

WHAT REPLACES THE OSTEON

Structure	What it is
Trabeculae	The open lattice of bony struts that forms spongy bone, aligned along lines of stress
No osteons	Spongy bone has no osteons. Its spaces hold red marrow

HOLD ONTO THIS

Compact bone solves the problem of strength with the osteon. Spongy bone solves the problem of weight with the trabecula. Same tissue, two answers, and you can tell them apart on a slide by looking for a central canal.

The four bone cells

CELL, JOB, AND WHERE IT COMES FROM

Cell	What it does	Origin or fate
Osteogenic cell	Divides to produce new osteoblasts	The stem cell of bone tissue
Osteoblast	Builds and secretes new bone matrix	Becomes an osteocyte once surrounded by matrix
Osteocyte	Maintains the bone matrix day to day	A mature osteoblast, sitting in a lacuna
Osteoclast	Breaks down and resorbs bone matrix	A large multinucleated cell from a blood-cell line, unlike the other three

HOLD ONTO THIS

Three of the four are one lineage: osteogenic becomes osteoblast becomes osteocyte. The osteoclast is the outsider, from a blood-cell line, and it is the only one that takes bone away.

Bone Growth and Formation

This section follows the structures and the sequence: the two ways bone forms, the two ways it grows, and the ordered zones of the growth plate. The chemistry belongs to physiology and is not covered here.

Ossification, the two routes that form bone

WHAT EACH ROUTE BUILDS

Route	What it builds	In brief
Intramembranous	The flat bones of the skull	Bone forms directly within a connective tissue membrane
Endochondral	Most bones of the body	Bone replaces a hyaline cartilage model

The two ways a long bone grows

The **epiphyseal plate** is the disc of hyaline cartilage where lengthening happens. When growth ends, the plate closes and leaves the **epiphyseal line**.

DIRECTION AND LOCATION

Type of growth	Direction	Where it happens
Interstitial growth	The bone grows longer	At the epiphyseal plate
Appositional growth	The bone grows wider	At the outer surface, beneath the periosteum

Zones of the growth plate

Five zones in order, from the epiphysis at the top to the diaphysis at the bottom.

1. **Zone of resting cartilage.** Anchors the growth plate to the epiphysis.
2. **Zone of proliferation.** Chondrocytes divide and stack into columns.
3. **Zone of hypertrophy.** The chondrocytes enlarge.
4. **Zone of calcification.** The cartilage matrix calcifies.
5. **Zone of ossification.** Cartilage is replaced by bone next to the diaphysis.

Classification: the supportive connective tissues

Bone histology fits inside one family. This map ties the chapter together: cartilage and the two kinds of bone, each with its structural unit.

THE FAMILY, WITH EACH STRUCTURAL UNIT

Tissue	Its members or its unit
Cartilage	Hyaline, elastic, fibrocartilage
Compact bone	Built from osteons: central canal, lamellae, lacunae, canaliculi
Spongy bone	Built from trabeculae, with red marrow in the spaces

BONE STRUCTURE SHOWS UP IN EVERY X-RAY

The growth plate is cartilage, so on a child's X-ray it looks like a gap. A fracture that runs through the plate can slow or stop a bone from growing, which is why pediatric fractures are described by how they cross it. The epiphyseal line is the closed plate, and its presence tells a clinician the skeleton is mature. Compact bone is built from osteons aligned with the lines of stress, so a bone is strongest along its long axis. That is why a twisting force, off that axis, is the one that tends to break it.

CHAPTER 2

The Skull

AXIAL SKELETON

The skull is 22 bones in two groups, the cranium that encloses the brain and the facial skeleton that frames the face. This chapter names every bone, the sutures and fontanelles, the key markings, the cavities, and the major foramina.

BY THE END

1. Distinguish the cranial bones from the facial bones and name the bones in each group.
2. Identify the significant markings on each cranial and facial bone, and the major sutures and fontanelles.
3. Identify the bones that form the orbit, the nasal cavity, and the hard palate, and locate the paranasal sinuses.
4. Name the major foramina of the skull and the structure each one transmits.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Axial Skeleton, Skull

drsrennie-stack.github.io/new-build-bio4-solano/axial-concept-videos.html

The Skull, an Overview

The skull has 22 bones. The eight cranial bones enclose the brain, and the fourteen facial bones frame the face. The hyoid bone of the neck is associated with the skull but is counted separately.

ORIENTATION TERMS

Term	What it is
The skull	The bony framework of the head, 22 bones, plus the associated hyoid bone
Cranium	The eight cranial bones that enclose and protect the brain
Facial skeleton	The fourteen facial bones that frame the face and hold the teeth
Calvaria	The domed roof of the cranium, the skullcap
Cranial base	The floor of the cranium, shaped into anterior, middle, and posterior cranial fossae

The Cranial Bones

Eight bones form the neurocranium, the braincase. The frontal, occipital, sphenoid, and ethmoid are single. The parietal and temporal bones are paired.

THE EIGHT AT A GLANCE

Bone	Number	Region it forms
Frontal bone	Single	The forehead, the roof of the orbits, and the floor of the anterior cranial fossa
Parietal bones	Paired	The superior and lateral walls of the cranium
Temporal bones	Paired	The inferior lateral walls and part of the cranial base. House the ear
Occipital bone	Single	The posterior wall and much of the cranial base
Sphenoid bone	Single	The central wedge of the cranial base. Articulates with every other cranial bone
Ethmoid bone	Single	The deepest cranial bone, between the orbits. Forms part of the nasal cavity

Frontal bone

MARKINGS

Marking	What it is
Squamous part	The broad plate of the forehead
Supraorbital margin	The ridge above each orbit
Supraorbital foramen	The opening in that margin. Passes the supraorbital nerve and vessels
Glabella	The smooth area between the eyebrow ridges
Orbital plates	The paired plates that roof the orbits and floor the anterior cranial fossa
Frontal sinus	The air space within the bone, above the orbits

Parietal bones

MARKINGS

Marking	What it is
The cranial roof	The pair form most of the superior and lateral walls of the cranium
Parietal eminence	The rounded high point of each bone, the skull's widest measure
Suture borders	They meet at the sagittal suture and form the coronal, lambdoid, and squamous sutures with neighbouring bones

Temporal bones

MARKINGS

Marking	What it is
Squamous part	The flat region forming the lower lateral wall of the skull
Zygomatic process	The bar that joins the zygomatic bone to complete the zygomatic arch
Mandibular fossa	The socket that receives the mandible, forming the temporomandibular joint
External acoustic meatus	The opening of the ear canal

Marking	What it is
Mastoid process	The bump behind the ear, a muscle attachment, filled with air cells
Styloid process	The slender spike below the ear that anchors tongue and neck muscles
Petrous part	The dense wedge of the cranial base that houses the middle and inner ear
Internal acoustic meatus	The canal carrying the nerves for hearing and facial movement
Stylomastoid foramen	The exit of the facial nerve, between the styloid and mastoid processes

Occipital bone

MARKINGS

Marking	What it is
Foramen magnum	The large central opening for the spinal cord
Occipital condyles	The paired knobs beside the foramen magnum that articulate with the atlas
External occipital protuberance	The midline bump felt at the back of the head
Hypoglossal canal	The canal passing the hypoglossal nerve, cranial nerve XII, to the tongue muscles
Foramen magnum contents	The medulla oblongata, the vertebral arteries, the anterior and posterior spinal arteries, the spinal accessory nerve (CN XI), and the alar and apical ligaments of the dens
Atlanto-occipital joint	The occipital condyles meeting the superior articular facets of the atlas, C1. The nodding joint that says yes

MNEMONIC

For the contents of the foramen magnum: VAMPires Sing AT Midnight. Vertebral arteries, Anterior and posterior spinal arteries, Membranes and ligaments, Spinal accessory nerve, Medulla.

Sphenoid bone

MARKINGS

Marking	What it is
Body	The central cube of the bone. Contains the sphenoid sinus
Sella turcica	The saddle on the body whose deep seat cradles the pituitary gland
Greater wings	The broad lateral wings forming part of the cranial floor, the orbit, and the lateral skull wall
Lesser wings	The smaller anterior wings forming part of the anterior cranial fossa
Pterygoid processes	The downward projections that anchor the chewing muscles
Optic canal	Passes the optic nerve from the eye
Superior orbital fissure	Passes the nerves that move the eye

Ethmoid bone

MARKINGS

Marking	What it is
Crista galli	The upright ridge where the falx cerebri, a fold of the brain covering, attaches
Cribriform plate	The perforated horizontal plate that the olfactory nerve filaments pass through
Perpendicular plate	The vertical plate forming the upper part of the nasal septum
Lateral masses	The side portions, which contain the ethmoidal air cells
Superior and middle conchae	The scroll-shaped projections into the nasal cavity

The Facial Bones

Fourteen bones form the viscerocranium, the facial skeleton. Six are paired, and two, the vomer and the mandible, are single.

THE FOURTEEN AT A GLANCE

Bone	Number	What it forms
Maxillae	Paired	The upper jaw. Form the anterior hard palate and hold the upper teeth
Palatine bones	Paired	The posterior hard palate and part of the orbit
Zygomatic bones	Paired	The cheekbones
Nasal bones	Paired	The bridge of the nose
Lacrimal bones	Paired	The medial wall of the orbit, grooved for the tear duct
Inferior nasal conchae	Paired	Scroll-shaped bones on the lateral wall of the nasal cavity
Vomer	Single	The inferior part of the nasal septum
Mandible	Single	The lower jaw, the only freely movable bone of the skull

The maxillae

MARKINGS

Marking	What it is
Body	The central mass of the bone. Contains the maxillary sinus
Alveolar process	The ridge that holds the upper teeth
Palatine process	The shelf forming the anterior part of the hard palate
Frontal process	The upward projection that meets the frontal bone
Zygomatic process	The lateral projection that meets the cheekbone
Infraorbital foramen	The opening below the orbit. Passes the infraorbital nerve to the face

The mandible

MARKINGS

Marking	What it is
Body	The horizontal, U-shaped portion that holds the lower teeth
Ramus	The vertical portion on each side
Condylar process	The posterior process whose head meets the temporal bone at the jaw joint
Coronoid process	The anterior process where the temporalis chewing muscle attaches
Mandibular notch	The gap between the condylar and coronoid processes
Alveolar process	The ridge that holds the lower teeth
Mandibular foramen	The opening on the inner ramus. The entry for the nerve to the lower teeth, and the target for a dental block

Sutures and Fontanelles

The major sutures

A suture is an immovable fibrous joint between skull bones.

BONES JOINED AND LOCATION

Suture	Bones it joins	Location
Coronal suture	The frontal bone and the two parietal bones	Runs across the top of the skull, behind the forehead
Sagittal suture	The two parietal bones	The midline of the skull roof
Lambdoid suture	The parietal bones and the occipital bone	The back of the skull
Squamous suture	A parietal bone and a temporal bone	The lower side of the skull, one on each side
Pterion	The frontal, parietal, temporal, and sphenoid bones	The H-shaped junction on the side of the skull. The thinnest part

The fontanelles

Fontanelles are fibrous membrane gaps between the bones of an infant skull. They let the skull compress during birth and grow with the brain.

LOCATION AND CLOSURE

Fontanelle	Also called	Location	Closes by
Anterior fontanelle	The largest fontanelle	Where the frontal and parietal bones meet	About 18 to 24 months
Posterior fontanelle	The occipital fontanelle	Where the parietal bones meet the occipital bone	About 2 months
Anterolateral fontanelles	The sphenoidal fontanelles	On the side of the skull, one on each side	About 3 months
Posterolateral fontanelles	The mastoid fontanelles	Behind the ear, one on each side	Begin at 1 to 2 months, complete by about 12 months

Cavities and Special Features

The orbit and nasal cavity

WHAT BUILDS EACH SPACE

Structure	What it is
The orbit	The bony eye socket, built from seven bones: frontal, sphenoid, zygomatic, maxilla, lacrimal, ethmoid, and palatine
Nasal cavity	The space behind the nose, with a bony roof, floor, lateral walls, and septum
Nasal septum	The partition of the nasal cavity. The vomer plus the perpendicular plate of the ethmoid plus septal cartilage
Hard palate	The bony roof of the mouth. The maxillae in front and the palatine bones behind

Paranasal sinuses

Air-filled cavities that lighten the skull and drain into the nasal cavity.

FOUR SINUSES, BY THE BONE EACH OCCUPIES

Sinus	Bone it occupies	Location
Frontal sinus	The frontal bone	Above the orbits
Ethmoidal sinuses	The ethmoid bone	The ethmoid air cells, between the orbits
Sphenoidal sinus	The sphenoid bone	Within the body of the sphenoid
Maxillary sinuses	The maxillae	The largest sinuses, in the cheek region

The hyoid bone

THE BONE THAT TOUCHES NO OTHER BONE

Feature	What it is
Hyoid bone	A U-shaped bone in the anterior neck, below the mandible
Role	Anchors the tongue and several muscles of the neck and larynx
Unique feature	It does not articulate with any other bone. It is slung by ligaments from the styloid processes

Major Foramina

A foramen is an opening that lets a nerve or a vessel pass through bone.

BONE PIERCED AND WHAT PASSES THROUGH

Foramen	Bone	What passes through
Foramen magnum	Occipital bone	The spinal cord and the vertebral arteries
Optic canal	Sphenoid bone	The optic nerve, cranial nerve II
Superior orbital fissure	Sphenoid bone	Nerves that supply the eye, including those that move it
Foramen ovale	Sphenoid bone	A branch of the trigeminal nerve to the lower jaw
Cribriform foramina	Ethmoid bone	The olfactory nerve filaments, cranial nerve I
Internal acoustic meatus	Temporal bone	The nerves for hearing and for facial movement
Carotid canal	Temporal bone	The internal carotid artery
Jugular foramen	Between the temporal and occipital bones	The internal jugular vein
Mental foramen	Mandible	The nerve and vessels to the chin and lower lip

WHERE THE SKULL IS THIN, AND WHERE IT HAS ONLY ONE DOOR

The pterion, the H-shaped junction of the frontal, parietal, temporal, and sphenoid bones, is the thinnest part of the skull, and the middle meningeal artery runs just beneath it. A blow there can tear the artery and bleed into the skull. The paranasal sinuses drain into the nasal cavity through narrow openings, so a head cold can spread into them as sinusitis, and the maxillary sinus sits so close to the upper teeth that a tooth infection can reach it. A newborn is checked at the anterior fontanelle: a sunken fontanelle can signal dehydration, a bulging one can signal raised pressure inside the skull. The foramen magnum is the one large opening between the brain and the spinal cord, so when pressure builds inside the rigid skull, brain tissue can be forced down through it. That is why rising intracranial pressure is an emergency.

CHAPTER 3

The Vertebral Column and Thoracic Cage

AXIAL SKELETON

The vertebral column is the central axis of the body, and the thoracic cage is built onto it. This chapter covers the regions and curvatures, a typical vertebra, each region of vertebrae in detail, and the sternum and ribs.

BY THE END

1. Name the five regions of the vertebral column and the number of vertebrae in each.
2. Identify the parts of a typical vertebra and the structures between adjacent vertebrae.
3. Tell a cervical, thoracic, and lumbar vertebra apart from a single feature.
4. Name the four curvatures and sort them as primary or secondary.
5. Identify the parts of the sternum and classify the ribs by anterior attachment.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Axial Skeleton, Spine

drsrennie-stack.github.io/new-build-bio4-solano/axial-concept-videos.html

The Vertebral Column, an Overview

Early in development the column has 33 vertebrae. In the adult, 5 sacral and 4 coccygeal vertebrae fuse, leaving 26 bones: 24 separate vertebrae in three regions, plus the sacrum and the coccyx.

THE FIVE REGIONS, SUPERIOR TO INFERIOR

Region	Vertebrae	Description
Cervical region	7, C1 to C7	The neck
Thoracic region	12, T1 to T12	The chest. Each vertebra articulates with a pair of ribs
Lumbar region	5, L1 to L5	The lower back
Sacrum	5, fused	A single triangular bone wedged between the hip bones
Coccyx	4, fused	The tailbone

Intervertebral discs are fibrocartilage pads between adjacent vertebral bodies.

The Curvatures

The adult spine has four front-to-back curves that act like a spring, giving the column strength, flexibility, and balance. Primary curvatures are present at birth. Secondary curvatures develop with the milestones of infancy.

TYPE AND TIMING

Curvature	Type	When it develops
Cervical curvature	Secondary	Develops as an infant holds up its head
Thoracic curvature	Primary	Present from birth
Lumbar curvature	Secondary	Develops as a child stands and walks
Sacral curvature	Primary	Present from birth

HOLD ONTO THIS

The primary curves are the two you were born with, and they both curve the same way as the fetal C shape. The secondary curves are the two you earned, one by lifting your head and one by standing up.

A Typical Vertebra

Parts of a vertebra

FROM THE BODY OUTWARD

Part	What it is
Body	The thick, weight-bearing anterior portion. Also called the centrum
Vertebral arch	The bony ring projecting from the back of the body
Pedicles	The two short bars connecting the arch to the body
Laminae	The two flat plates that complete the arch behind
Vertebral foramen	The opening enclosed by the body and arch. The spinal cord runs through it
Spinous process	The single projection pointing backward from the arch
Transverse processes	The two projections pointing laterally from the arch
Articular processes	The superior and inferior pairs that join one vertebra to the next

Between the vertebrae

THE SPACES AND THE DISCS

Structure	What it is
Vertebral canal	The stacked vertebral foramina, the channel that holds the spinal cord
Intervertebral foramina	The gaps between stacked vertebrae where the spinal nerves exit
Intervertebral disc	A fibrocartilage pad between two vertebral bodies, a shock absorber
Anulus fibrosus	The tough outer ring of the disc
Nucleus pulposus	The soft, gel-like core of the disc

Regional Vertebrae

Each region has a vertebra shaped for its job. One feature is enough to place a loose vertebra.

BODY, SPINOUS PROCESS, AND THE GIVEAWAY FEATURE

Region	Body	Spinous process	Distinctive feature
Cervical, C1 to C7	Small and oval, the smallest vertebrae	Short, often bifid or forked	Transverse foramina in the transverse processes, which pass the vertebral arteries
Thoracic, T1 to T12	Heart-shaped	Long, angled sharply downward	Costal facets on the body and transverse processes where the ribs articulate
Lumbar, L1 to L5	Thick and kidney-shaped, the largest vertebrae	Short and blunt	Massive build for weight bearing. No costal facets and no transverse foramina

The specialized cervical vertebrae

THE TWO THAT ARE NOT TYPICAL

Structure	What it is
Atlas (C1)	A ring with no body and no spinous process. It holds up the skull
Axis (C2)	Bears the dens, an upward bony peg
Dens	The odontoid process, the peg of the axis that the atlas pivots around for the no rotation
Atlanto-occipital joint	Where the atlas meets the occipital condyles. The joint for the yes nod
Atlanto-axial joint	Where the atlas pivots on the dens of the axis. The joint for the no head rotation
Vertebra prominens (C7)	A long spinous process you can feel at the base of the neck

Sacrum and coccyx

THE FUSED REGIONS

Structure	What it is
Sacrum	Five fused vertebrae forming a triangular bone wedged between the hip bones
Sacral promontory	The anterior ridge of the first sacral vertebra
Median sacral crest	The fused spinous processes along the posterior surface
Sacral foramina	The openings that pass the sacral nerves
Sacral canal	The continuation of the vertebral canal through the sacrum
Sacral hiatus	The opening at the inferior end of the sacral canal
Auricular surface	The ear-shaped surface that joins the hip bone at the sacroiliac joint
Coccyx	The fused tailbone, an attachment point for pelvic ligaments and muscles

The Thoracic Cage

The thoracic cage is the sternum in front, the ribs on the sides, and the thoracic vertebrae behind. It shields the heart and lungs.

The sternum

THREE FUSED PARTS AND THEIR LANDMARKS

Part	What it is
Sternum	The breastbone, a flat bone of three fused parts
Manubrium	The superior part. Articulates with the clavicles and the first ribs
Body	The long middle part of the sternum
Xiphoid process	The small inferior tip, a muscle attachment. Cartilage that ossifies with age
Jugular notch	The central dip at the top of the manubrium, easily felt
Sternal angle	The ridge where the manubrium meets the body. A landmark level with the second rib

The ribs

PARTS OF A RIB

Part	What it is
Ribs	Twelve pairs of flat, curved bones forming the walls of the cage
Head	The posterior end of a rib. Articulates with the vertebral bodies
Tubercle	The knob near the head. Articulates with a transverse process
Costal cartilage	The hyaline cartilage that connects a rib toward the sternum

THE THREE GROUPS, BY ANTERIOR ATTACHMENT

Rib group	Pairs	Anterior attachment
True ribs	1 to 7	Attach to the sternum directly, each by its own costal cartilage
False ribs	8 to 12	Do not reach the sternum directly. Pairs 8 to 10 join the costal cartilage of the rib above
Floating ribs	11 and 12	No anterior attachment at all. A subset of the false ribs

THE SPINE PROTECTS A CORD, THE CAGE PROTECTS A PUMP

The vertebral canal carries the spinal cord, so a vertebral fracture is dangerous for what runs inside it, not only for the bone. A herniated disc occurs when the soft nucleus pulposus pushes through a torn anulus fibrosus and presses on a spinal nerve at an intervertebral foramen, which is why the pain travels down a limb. The dens of the axis is held against the atlas by a ligament, and a sharp blow can fracture it, a serious injury so close to the brainstem. The sternal angle is a landmark a clinician feels for, sitting at the level of the second rib, the starting point for counting ribs, and the xiphoid process guides hand position for chest compressions in CPR. Abnormal curvatures have names: scoliosis is a sideways curve, kyphosis is an exaggerated thoracic curve, and lordosis is an exaggerated lumbar curve.

CHAPTER 4

The Upper Extremity

APPENDICULAR SKELETON

The pectoral girdle and the bones of the arm, forearm, and hand. This chapter also carries the bone markings vocabulary, which repeats on every bone for the rest of the course.

BY THE END

1. Name the two bones of the pectoral girdle and explain how the girdle attaches the upper limb to the axial skeleton.
2. Trace the bones of the upper limb from shoulder to fingertips.
3. Tell the radius from the ulna by side and by the markings at each end.
4. Name the three groups of hand bones and how many bones are in each.
5. Use the bone markings vocabulary to classify any landmark as a projection, a depression, or an opening.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Appendicular Skeleton, Upper

drsrennie-stack.github.io/new-build-bio4-solano/appendicular-concept-videos.html

Bone Markings Vocabulary

Learn this before the landmarks. Every marking is a projection, a depression, or an opening, and these terms repeat on every bone you will study.

TYPE, DESCRIPTION, EXAMPLE

Marking	Type	Description	Example
Process	Projection	Any bony prominence	Mastoid process
Tubercle	Projection	Small rounded bump	Greater tubercle of the humerus
Tuberosity	Projection	Large rough projection	Deltoid tuberosity
Head	Projection	Bony expansion carried on a neck	Head of the humerus
Epicondyle	Projection	Raised area above a condyle	Medial epicondyle of the humerus
Facet	Articular surface	Smooth, nearly flat surface	Articular facets of vertebrae
Fossa	Depression	Shallow basin	Olecranon fossa
Sulcus	Depression	A groove	Intertubercular sulcus
Foramen	Opening	Hole for vessels or nerves	Nutrient foramen

The Upper Extremity, an Overview

The appendicular skeleton is the bones of the limbs and the girdles that attach them to the axial skeleton.

FOUR PARTS, PROXIMAL TO DISTAL

Part	Region it spans	Bones
Pectoral girdle	Attaches the upper limb to the trunk	Clavicle and scapula
Arm	Shoulder to elbow	Humerus
Forearm	Elbow to wrist	Radius and ulna
Hand	Wrist to fingertips	Carpals, metacarpals, and phalanges

The Pectoral Girdle

Clavicle

MARKINGS

Feature	What it is
Clavicle	The collarbone, an S-shaped bone across the top of the thorax
Sternal end	The medial end. Articulates with the manubrium of the sternum
Acromial end	The lateral end. Articulates with the acromion of the scapula
The only bony link	The clavicle is the sole bony attachment of the upper limb to the axial skeleton

Scapula

MARKINGS

Feature	What it is
Scapula	The shoulder blade, a flat triangular bone on the posterior thorax
Borders	The superior, medial, and lateral borders of the triangle
Spine	The ridge running across the posterior surface
Acromion	The flat process at the tip of the spine, the point of the shoulder
Coracoid process	The hook-like anterior process, a muscle attachment
Glenoid cavity	The shallow socket that receives the head of the humerus
Supraspinous fossa	The hollow above the spine
Infraspinous fossa	The hollow below the spine
Subscapular fossa	The hollow on the anterior surface

Joints of the pectoral girdle

THREE JOINTS

Joint	What meets what
Sternoclavicular joint	Sternal end of the clavicle with the manubrium. The only joint linking the upper limb to the axial skeleton
Acromioclavicular joint	Acromial end of the clavicle with the acromion of the scapula
Glenohumeral joint	Head of the humerus in the glenoid cavity. The shoulder joint

The Arm and Forearm

Humerus

MARKINGS, PROXIMAL TO DISTAL

Marking	What it is
Humerus	The single bone of the arm, from shoulder to elbow
Head	The rounded proximal end that fits into the glenoid cavity
Anatomical neck	The narrow groove just below the head
Surgical neck	The narrowing below the tubercles. A common fracture site
Greater tubercle	The lateral bump near the head, a muscle attachment
Lesser tubercle	The anterior bump near the head, a muscle attachment
Intertubercular sulcus	The groove between the tubercles that holds a biceps tendon
Deltoid tuberosity	The roughened ridge on the shaft where the deltoid muscle attaches
Capitulum	The lateral knob at the distal end. Articulates with the radius
Trochlea	The medial spool at the distal end. Articulates with the ulna
Medial epicondyle	The medial projection above the trochlea
Lateral epicondyle	The lateral projection above the capitulum
Olecranon fossa	The deep posterior pit that receives the ulna when the elbow straightens

HOLD ONTO THIS

Capitulum sits lateral and meets the radius. Trochlea sits medial and meets the ulna. Both pairings hold in anatomical position, so fix the position first and the pairing follows.

The forearm bones, radius and ulna

SIDE AND MARKINGS AT EACH END

Bone	Side of the forearm	Proximal (elbow) end	Distal (wrist) end
Radius	Lateral, the thumb side	Disc-shaped head articulates with the capitulum. Neck just below. Radial tuberosity, a muscle attachment, below the neck	Styloid process on the lateral side. Ulnar notch on the medial side receives the head of the ulna, forming the distal radioulnar joint
Ulna	Medial, the little-finger side	Olecranon, the point of the elbow. Coronoid process in front. Trochlear notch grips the trochlea of the humerus. Radial notch on the lateral side receives the head of the radius, forming the proximal radioulnar joint	Styloid process on the medial side

Grooves, nerves, and the joining membrane

WHAT RUNS BETWEEN AND ALONG THE BONES

Structure	What it is
Radial groove	An oblique groove on the posterior shaft of the humerus. The radial nerve runs in it
Ulnar nerve at the medial epicondyle	The nerve passes just behind the medial epicondyle, where it can be palpated. The funny bone
Interosseous membrane	A fibrous sheet, a syndesmosis, spanning the shafts of the radius and ulna. A muscle attachment that also binds the two bones
Three radioulnar joints	Proximal (radial head and radial notch), distal (ulnar head and ulnar notch), and the interosseous membrane between the shafts

The Hand

Carpals, the wrist

EIGHT BONES IN TWO ROWS

Group	Bones
Carpals	Eight small bones arranged in two rows, forming the wrist
Proximal row	Scaphoid, lunate, triquetrum, and pisiform
Distal row	Trapezium, trapezoid, capitate, and hamate

MNEMONIC

So Long To Pinky, Here Comes The Thumb.

Metacarpals and phalanges

THE PALM AND THE DIGITS

Structure	What it is
Metacarpals	Five bones forming the palm, numbered 1 to 5 starting at the thumb
Phalanges	The finger bones, fourteen in each hand
Thumb	The pollex. Has two phalanges, a proximal and a distal
Fingers	Each of the other four has three phalanges: proximal, middle, and distal

The carpal tunnel

ROOF, CONTENTS, AND WHAT GOES WRONG

Structure	What it is
Flexor retinaculum	A strong fibrous band roofing the carpal bones. The space beneath it is the carpal tunnel
Contents	The long flexor tendons of the fingers and thumb, and the median nerve
Carpal tunnel syndrome	Narrowing of the tunnel compresses the median nerve
Clinically notable carpals	Capitate is the largest. Scaphoid is the most frequently fractured. Lunate is the most frequently dislocated

THE BONES THAT BREAK WHEN YOU FALL ON AN OUTSTRETCHED HAND

The clavicle is the most commonly broken bone in the body, usually in its middle third. It is the slender strut that takes the force when you fall onto a shoulder or an outstretched arm. The surgical neck of the humerus fractures often, and the axillary nerve runs close by, so a break there is checked for nerve injury. The scaphoid is the carpal bone most often fractured, again from a fall on an outstretched hand, and its blood supply is poor, so a missed scaphoid fracture may fail to heal. Tenderness in the anatomical snuffbox, the hollow at the base of the thumb, is the clue. The ulnar nerve passes behind the medial epicondyle of the humerus, the spot people call the funny bone, which is why a knock there sends a jolt to the little finger.

CHAPTER 5

The Lower Extremity

APPENDICULAR SKELETON

The pelvic girdle that carries the body's weight to the legs, and the bones of the thigh, leg, and foot, each broken out with its key markings.

BY THE END

1. Name the three bones that fuse to form the hip bone, and describe the pelvis.
2. Identify the key markings of the hip bone, including the acetabulum.
3. Identify the key markings of the femur, patella, tibia, and fibula.
4. Name the bones of the foot, including the seven tarsal bones and the arches.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Appendicular Skeleton, Lower

drsrennie-stack.github.io/new-build-bio4-solano/appendicular-concept-videos.html

The Lower Extremity, an Overview

The lower limb is built heavier than the upper limb because it carries the body's weight.

FOUR PARTS, PROXIMAL TO DISTAL

Part	Region it spans	Bones
Pelvic girdle	Attaches the lower limb to the axial skeleton	The two hip bones
Thigh	Hip to knee	Femur, and the patella at the knee
Leg	Knee to ankle	Tibia and fibula
Foot	Ankle to toes	Tarsals, metatarsals, and phalanges

The Pelvic Girdle

Each hip bone, the os coxae, begins as three separate bones that fuse at the acetabulum: the ilium, the ischium, and the pubis.

Ilium

MARKINGS

Marking	What it is
Ilium	The large, fan-shaped superior part of the hip bone
Iliac crest	The curved upper ridge, the rim you rest your hands on
Anterior superior iliac spine	The prominent point at the front of the crest, a palpable landmark
Posterior superior iliac spine	The point at the back of the iliac crest
Iliac fossa	The smooth, concave inner surface
Auricular surface	The ear-shaped surface that joins the sacrum at the sacroiliac joint
Greater sciatic notch	The deep notch on the posterior border. The sciatic nerve, the largest nerve in the body, passes through it

Ischium and pubis

MARKINGS

Marking	What it is
Ischium	The posteroinferior part of the hip bone
Ischial tuberosity	The strong, roughened knob you sit on
Ischial spine	The pointed projection above the tuberosity
Pubis	The anteromedial part of the hip bone
Pubic symphysis	The cartilage joint where the two pubic bones meet in front
Pubic crest	The ridge along the upper border of the pubis
Pubic rami	The superior and inferior bars of bone extending from the body

The acetabulum and the pelvis

WHERE THE THREE BONES MEET, AND THE PELVIS AS A WHOLE

Structure	What it is
Acetabulum	The deep cup where the three bones meet, the socket for the head of the femur
Obturator foramen	The large opening framed by the ischium and the pubis
The bony pelvis	The two hip bones together with the sacrum and the coccyx
True and false pelvis	The false pelvis is the wide flare above the pelvic brim. The true pelvis is the narrower ring below it
Sex differences	The female pelvis is wider and shallower, with a broader pubic angle and a larger outlet, adapted for childbirth

The Thigh and Knee

Femur

MARKINGS, PROXIMAL TO DISTAL

Marking	What it is
Femur	The thigh bone, the longest and strongest bone in the body
Head	The ball at the proximal end that fits into the acetabulum
Fovea capitis	The small pit on the head that anchors a ligament to the acetabulum
Neck	The narrow region just below the head, a common fracture site
Greater trochanter	The large lateral projection, a major muscle attachment
Lesser trochanter	The smaller projection on the medial side
Linea aspera	The rough ridge down the posterior shaft, a muscle attachment line
Medial and lateral condyles	The rounded distal knobs that articulate with the tibia

Marking	What it is
Medial and lateral epicondyles	The projections just above the condyles
Intercondylar fossa	The notch between the condyles on the posterior surface

Patella

THE SESAMOID AT THE KNEE

Feature	What it is
Patella	The kneecap, a sesamoid bone formed within the quadriceps tendon
Base	The broad superior border of the patella
Apex	The pointed inferior tip

The Leg

TIBIA AND FIBULA COMPARED

Bone	Side and weight	Proximal (knee) end	Distal (ankle) end
Tibia	Medial, the shin bone, the weight-bearing bone of the leg	Medial and lateral condyles articulate with the femur. Tibial tuberosity on the front anchors the patellar ligament. Sharp anterior border felt as the shin	Medial malleolus, the medial bump of the ankle
Fibula	Lateral, slender, does not bear body weight	Head, the knob at the proximal end	Lateral malleolus, the lateral bump of the ankle

HOLD ONTO THIS

The tibia carries the weight and the fibula does not. That one fact tells you which malleolus is which, why a length of fibula can be harvested as a graft, and why a tibial fracture is the more serious of the two.

The Foot

Tarsals, the ankle

SEVEN BONES

Structure	What it is
Tarsals	Seven bones forming the ankle and the back of the foot
Talus	The tarsal that articulates with the tibia and fibula and receives the body's weight
Calcaneus	The heel bone, the largest tarsal
Other tarsals	The navicular, the cuboid, and the three cuneiform bones

Metatarsals, phalanges, and arches

THE SOLE AND THE DIGITS

Structure	What it is
Metatarsals	Five bones forming the sole, numbered 1 to 5 from the great toe
Phalanges	The toe bones, fourteen in each foot
Great toe	The hallux. Has two phalanges, the other toes have three each
Arches of the foot	The medial and lateral longitudinal arches and the transverse arch, which spread body weight

THE SKELETON THAT CARRIES YOUR WEIGHT

A broken hip is usually a fracture of the femoral neck, common in older adults whose bone has thinned with osteoporosis. The neck is narrow and angled, and the blood supply to the femoral head runs close by, so the fracture can threaten the head itself. The anterior superior iliac spine and the iliac crest are landmarks a clinician feels through the skin, used to locate injection sites and to check pelvic alignment. The tibia carries body weight and the fibula does not, which is why a length of fibula can be taken as a bone graft, and the lateral malleolus of the fibula is the common ankle fracture site, often from a rolled ankle. The arches of the foot spread weight across the sole and spring back with each step. When the medial longitudinal arch drops, the result is flat feet.

CHAPTER 6

Articulations and Joints

JOINTS

A joint is where two or more bones meet. This chapter covers both classifications, the three structural classes in detail, the synovial joint and its accessory structures, the movements, the shoulder and the knee, and the common disorders.

BY THE END

1. Classify joints by structure into fibrous, cartilaginous, and synovial, and by function into synarthrosis, amphiarthrosis, and diarthrosis.
2. Name the six types of synovial joint and give a body example of each.
3. Use the parts of a synovial joint to explain how it allows movement with little friction.
4. Name the movements that occur at synovial joints.
5. Describe the shoulder and the knee, and explain what makes each one vulnerable.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Joints

drsrennie-stack.github.io/new-build-bio4-solano/joints-concept-videos.html

Classifying Joints

Joints are classified two ways: by structure and by function. Learn both, because a tagged joint can be asked either way.

STRUCTURAL CLASSIFICATION: CAVITY PRESENT, AND WHAT HOLDS THE BONES

Structural class	Synovial cavity	Connective tissue holding the bones
Fibrous joint	None	Dense connective tissue rich in collagen fibers
Cartilaginous joint	None	Cartilage, hyaline or fibrocartilage
Synovial joint	Present	The dense irregular connective tissue of an articular capsule

FUNCTIONAL CLASSIFICATION: DEGREE OF MOVEMENT

Functional class	Degree of movement	Examples
Synarthrosis	Immovable	Sutures of the skull, gomphoses
Amphiarthrosis	Slightly movable	Syndesmoses, symphyses
Diarthrosis	Freely movable	All synovial joints

Fibrous Joints

No synovial cavity. The bones are held closely by dense connective tissue, and they permit little or no movement.

STRUCTURE, MOBILITY, EXAMPLES

Type	Structure	Mobility	Examples
Suture	Thin layers of dense irregular connective tissue, with interlocking edges that add strength and absorb shock	Immovable in adults, a synarthrosis. Slightly movable in infants and children, an amphiarthrosis	Only between the bones of the skull. The metopic suture is a frontal suture that persists past about age six
Syndesmosis	More dense connective tissue and more space than a suture, the bones united by a ligament	Slightly movable, an amphiarthrosis	The distal tibiofibular joint, united by the anterior tibiofibular ligament
Gomphosis	A cone-shaped peg held in a socket by the periodontal ligament	Immovable, a synarthrosis	A tooth in its bony socket
Interosseous membrane	A substantial sheet of dense connective tissue binding neighbouring long bones	Slightly movable, an amphiarthrosis	Between the radius and ulna, between the tibia and fibula
Synostosis	A joint where two separate bones fuse completely into one	Immovable	A suture that ossifies in adulthood

Cartilaginous Joints

No synovial cavity, and little or no movement.

THE TWO TYPES, BY CONNECTING CARTILAGE

Type	Connecting cartilage	Mobility	Examples
Synchondrosis	Hyaline cartilage	Immovable, a synarthrosis	The epiphyseal growth plate, which becomes a synostosis when bone replaces the cartilage. The joint between the first rib and the manubrium
Symphysis	A broad, flat disc of fibrocartilage	Slightly movable, an amphiarthrosis	The pubic symphysis, the manubriosternal joint, the intervertebral joints between vertebral bodies. All symphyses lie in the midline

Synovial Joint Structure

A synovial joint has a synovial cavity between the articulating bones and allows considerable movement. All synovial joints are diarthroses.

CARTILAGE AND CAPSULE

Structure	What it is
Articular cartilage	Hyaline cartilage covering the articulating bone surfaces. Smooth and slippery, it reduces friction and absorbs shock but does not bind the bones
Articular capsule	A sleeve-like capsule that surrounds the joint, encloses the synovial cavity, and unites the bones
Fibrous membrane	The outer layer of the capsule. Dense irregular connective tissue attached to the periosteum, strong and flexible
Ligaments	Thickened fibrous bundles of the fibrous membrane that resist strain and prevent dislocation
Synovial membrane	The inner layer of the capsule. Areolar connective tissue with elastic fibers
Articular fat pads	Accumulations of adipose tissue that may cushion the joint

SYNOVIAL FLUID, AND THE FOUR THINGS IT DOES

Feature	What it is or does
Synovial fluid	A viscous, clear, pale-yellow fluid of hyaluronic acid and interstitial fluid, secreted by the synovial membrane
Lubrication	Forms a thin film that reduces friction between the joint surfaces
Nourishment	Supplies oxygen and nutrients to, and removes wastes from, the avascular articular cartilage
Defence	Contains phagocytic cells that clear microbes and debris from normal wear
Warm-up effect	Gel-like at rest, it thins with movement, which is why warming up before exercise eases the joints

Accessory Structures of Synovial Joints

LIGAMENTS, DISCS, BURSAE, AND SUPPLY

Structure	What it is
Extracapsular ligaments	Ligaments that lie outside the articular capsule, as the collateral ligaments of the knee
Intracapsular ligaments	Ligaments inside the capsule but kept out of the synovial cavity by synovial folds, as the cruciate ligaments of the knee
Articular discs (menisci)	Crescent-shaped fibrocartilage pads between the bone surfaces. They improve fit, absorb shock, distribute weight, and spread synovial fluid
Labrum	A fibrocartilage rim around the socket of a ball-and-socket joint that deepens it. Found at the shoulder and hip
Bursae	Small sac-like structures filled with fluid, placed to reduce friction between skin, bone, muscle, tendon, or ligament
Tendon sheaths	Tube-shaped bursae that wrap a tendon where it passes through a tight space, as at the wrist and ankle
Nerve supply	The same nerves that supply the muscles moving the joint. They carry pain information and help adjust movement
Blood supply	Arteries near the joint branch into the capsule and ligaments, veins carry wastes away. The articular cartilage is fed instead by the synovial fluid

Movements at Synovial Joints

GLIDING AND ANGULAR MOVEMENTS

Movement	What it is
Gliding	Simple back-and-forth or side-to-side movement, no change in the angle between bones
Flexion	Decreases the angle between bones, usually in the sagittal plane
Extension	Increases the angle between bones, the reverse of flexion
Hyperextension	Extension that continues past the anatomical position
Lateral flexion	Bending the trunk or head sideways, in the frontal plane
Abduction	Movement away from the midline of the body
Adduction	Movement toward the midline of the body
Circumduction	Movement of the distal end of a part in a circle

ROTATION AND SPECIAL MOVEMENTS

Movement	What it is
Rotation	A bone turns around its own long axis. Medial rotation toward the midline, lateral rotation away
Elevation, depression	Lifting a body part upward, then lowering it

Movement	What it is
Protraction, retraction	Moving a body part forward, then back to position
Inversion, eversion	Turning the sole of the foot inward, then outward
Dorsiflexion, plantar flexion	Bending the foot up at the ankle, then pointing the toes down
Supination, pronation	Turning the palm to face anteriorly, then posteriorly
Opposition	Moving the thumb across the palm to touch the fingertips

The Six Types of Synovial Joints

Sorted by the shape of their surfaces, which sets the axes of movement. Uniaxial moves around one axis, biaxial around two, triaxial around three.

SHAPE, AXES, EXAMPLES

Joint type	Shape of the surfaces	Axes of movement	Examples
Plane joint	Flat or slightly curved surfaces that glide	Biaxial or triaxial gliding	The intercarpal and intertarsal joints
Hinge joint	A convex surface fits a concave one	Uniaxial, flexion and extension	The elbow, ankle, and interphalangeal joints
Pivot joint	A rounded surface fits a ring of bone and ligament	Uniaxial, rotation	The atlantoaxial and radioulnar joints
Condyloid joint	An oval projection fits an oval depression	Biaxial	The radiocarpal and metacarpophalangeal joints
Saddle joint	Each surface is both concave and convex	Biaxial	The carpometacarpal joint of the thumb
Ball-and-socket joint	A ball fits a cup	Triaxial	The shoulder and hip joints

Factors Affecting Range of Motion

WHAT SETS THE LIMITS

Factor	How it limits movement
Bone shape	The structure and fit of the articulating bones
Ligament tension	Taut ligaments restrict movement and direct the bones
Contact of soft parts	One body surface meeting another can limit movement
Hormones	Relaxin loosens the pubic symphysis and pelvic ligaments

The Shoulder Joint

The glenohumeral joint, a ball-and-socket joint between the head of the humerus and the glenoid cavity of the scapula. It is the most freely movable joint in the body, and the price of that mobility is instability.

STRUCTURE AND SUPPORTING LIGAMENTS

Structure	What it is
Type	A ball-and-socket joint, also called the glenohumeral or humeroscapular joint
Articulating bones	The head of the humerus and the glenoid cavity of the scapula
Why it moves so freely	The articular capsule is loose and the glenoid cavity is shallow relative to the large humeral head
Articular capsule	A thin, loose sac that envelops the joint, from the glenoid cavity to the anatomical neck of the humerus
Coracohumeral ligament	Runs from the coracoid process to the greater tubercle. Strengthens the superior part of the capsule
Glenohumeral ligaments	Three thickenings of the anterior capsule that stabilize the joint near the limits of motion
Transverse humeral ligament	A narrow band between the greater and lesser tubercles that holds the long head of the biceps tendon in place
Glenoid labrum	A rim of fibrocartilage around the glenoid cavity that slightly deepens and enlarges it
Bursae	Four are associated with the joint: the subscapular, subdeltoid, subacromial, and subcoracoid bursae
Movements	Flexion, extension, hyperextension, abduction, adduction, medial and lateral rotation, and circumduction
Source of stability	The ligaments add little strength. Most stability comes from the rotator cuff muscles
Rotator cuff	The supraspinatus, infraspinatus, teres minor, and subscapularis, which hold the humeral head in the glenoid cavity

MNEMONIC

The rotator cuff is SITS: Supraspinatus, Infraspinatus, Teres minor, Subscapularis.

The Knee Joint

The tibiofemoral joint, the largest and most complex joint in the body. A modified hinge joint, and the joint most often injured.

STRUCTURE, LIGAMENTS, AND MENISCI

Structure	What it is
Type	A modified hinge joint, the largest and most complex joint in the body
Three joints, one cavity	A medial and a lateral tibiofemoral joint between the femoral and tibial condyles, and a patellofemoral joint between the patella and the femur
Articular capsule	The capsule that encloses the joint
Patellar retinacula	The medial and lateral retinacula, fused quadriceps tendon and fascia, strengthen the anterior surface
Patellar ligament	The continuation of the quadriceps tendon from the patella to the tibial tuberosity. Strengthens the anterior surface

Structure	What it is
Oblique popliteal ligament	A broad ligament that strengthens the posterior surface of the joint
Arcuate popliteal ligament	Strengthens the lower lateral part of the posterior surface
Anterior cruciate ligament	An intracapsular ligament, crossing within the joint. Stops the tibia from sliding forward. Torn in roughly 70 percent of serious knee injuries
Posterior cruciate ligament	An intracapsular ligament, crossing within the joint. Stops the tibia from sliding backward. Important on stairs and inclines
Medial meniscus	The C-shaped fibrocartilage disc on the medial side of the joint
Lateral meniscus	The more circular, O-shaped disc on the lateral side
Transverse ligament	The band that joins the two menisci to each other

Knee injuries

WHAT GOES WRONG AND WHAT IT MEANS

Injury	What it is
Unhappy triad	A lateral blow to a planted knee that tears three structures at once: the tibial collateral ligament, the medial meniscus, and the anterior cruciate ligament
ACL and women	The anterior cruciate ligament is torn three to six times more often in women, from a narrower intercondylar space, a wider pelvis and greater knee angle, ligament-loosening hormones, and less protective muscle strength
Swollen knee	Immediate swelling is blood from torn vessels. Delayed swelling is excess synovial fluid
Dislocated knee	Displacement of the tibia relative to the femur, most often anterior. It can damage the popliteal artery
PRICE	The first-aid approach to a knee injury: protection, rest, ice, compression, elevation

Common Joint Disorders

DISORDER AND DESCRIPTION

Disorder	What it is
Double-jointed	Greater ligament flexibility and range of motion. The joints are less stable and dislocate more easily
Bursitis	Inflammation of a bursa, from repeated overuse, infection, or rheumatoid arthritis, with pain, swelling, and limited movement
Torn meniscus	A tear of the knee cartilage, common in athletes, repaired by arthroscopy with a fibre-optic camera
Rheumatism	Any painful disorder of the supporting structures, bones, ligaments, tendons, or muscles, not caused by infection or injury
Arthritis	A form of rheumatism with swollen, stiff, painful joints. The leading cause of physical disability in adults over 65

Disorder	What it is
Osteoarthritis	A progressive, degenerative loss of joint cartilage from aging and wear, in the large weight-bearing joints
Rheumatoid arthritis	An autoimmune disease that attacks the cartilage and joint linings, usually on both sides, inflaming the synovial membrane
Gouty arthritis	Sodium urate crystals deposit in joints, often the base of the big toe, eroding the cartilage with inflammation and pain

Classification: joints by structure

THE STRUCTURAL CLASSES AND THEIR SUBTYPES

Class	Subtypes and examples
Fibrous, joined by dense connective tissue	Sutures (skull), gomphoses (tooth sockets), syndesmoses (radius to ulna)
Cartilaginous, joined by cartilage	Synchondroses (epiphyseal plate, first rib), symphyses (pubic symphysis, intervertebral discs)
Synovial, joined across a fluid-filled cavity	Plane, hinge, pivot, condyloid, saddle, ball-and-socket

FREEDOM OF MOVEMENT HAS A COST

Synovial joints are freely movable, and that is exactly why they are the joints that get injured. A sprain is a stretched or torn ligament, the band that stabilizes a joint. Two arthritis types are worth telling apart: osteoarthritis strikes the large joints and is wear of the cartilage, while rheumatoid arthritis attacks the small joints first and actively destroys the cartilage and synovial membrane. Knowing the joint type tells you the movements to expect, and the ways the joint can fail.

PART B

Pre-work.

Four packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

1. **Bone Tissue**
2. **Axial Skeleton**
3. **Appendicular Skeleton**
4. **Joints**

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.

Pre-work · Bone Tissue

Bone Tissue and Microanatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 2 · 1 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the bone chapter into mini visuals. Seven chunks, each in the shape that fits it.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A slide of bone shows rings of matrix around a canal, with small cavities between the rings. Name the tissue and the unit.

Answer. Compact bone. The unit is the osteon.

Reasoning. Work from the structure you can actually see. Rings around a canal means concentric lamellae around a central canal, and that pairing exists only in compact bone. The small cavities are lacunae holding osteocytes. Spongy bone would show none of this, because it has no osteons at all, only trabeculae with marrow in the spaces. So the presence of a central canal is the single feature that settles it.

takes longer than three minutes you are copying instead of organizing.

1. **GRID**

   The three cartilages

   Hyaline, elastic, fibrocartilage down the side. Fibers, texture, and one location across the top. Add a fourth column for the joint each one turns up in.

2. **HUB**

   Gross anatomy of a long bone

   One long bone down the middle. Spoke out to diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and both marrows. Draw the bone, do not just list.

3. **HUB**

   The osteon

   Central canal at the hub. Spoke out to lamellae, lacunae, canaliculi, perforating canal, interstitial and circumferential lamellae. This is the single most tested structure in the chapter.

4. **SPLIT**

   Compact against spongy bone

   One split. Structural unit, what fills the spaces, and where in a long bone each is found. The giveaway on a slide goes on the branch.

5. **LADDER**

   The four bone cells

   Three rungs for the lineage, osteogenic to osteoblast to osteocyte, then the osteoclast off to one side because it comes from somewhere else. Write what each does beside its rung.

6. **SPLIT**

   Intramembranous against endochondral ossification

   What tissue each starts from, and which bones each builds. Two branches, no more.

DRAWING 1

A long bone, sectioned

BRAIN DUMP FIRST

Two minutes. Every part of a long bone, and which kind of bone tissue is where.

Draw a long bone cut down its length so the inside shows. Mark diaphysis, both epiphyses, metaphysis, medullary cavity, and where compact and spongy bone each sit. Add periosteum, endosteum, articular cartilage, the epiphyseal line, and label red and yellow marrow where each belongs.

In your notebook, head the page **M2–1 A long bone, sectioned** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at any level of your drawing and say which tissue is there and what it is doing.

Set A · Name it

1. The structural unit of compact bone is the ____________________.
2. The cell that resorbs bone matrix is the ____________________.
3. The cartilage capping the epiphysis where it meets another bone is ____________________.
4. The membrane lining the internal bone surfaces is the ____________________.
5. Name the two routes of ossification and one bone built by each.

____

Set B · Predict it

1. A child fractures through the epiphyseal plate. Predict the long-term consequence and say why.

____

____

2. An X-ray shows an epiphyseal line rather than a plate. What does that tell you about the person?

____

3. Canaliculi are blocked in a region of compact bone. Predict what happens to the osteocytes there.

____

____

4. Osteoclast activity outpaces osteoblast activity for years. Predict the effect on the skeleton and name one fracture it makes likely.

____

____

Set C · Tell it apart

1. A student says cartilage heals as fast as bone. Correct them in one sentence, using vascularity.

____

2. Distinguish appositional from interstitial growth in a long bone, by direction and by location.

____

____

7 LADDER

The five zones of the growth plate

Five rungs, resting cartilage at the top by the epiphysis, ossification at the bottom by the diaphysis. Write what the chondrocyte is doing at each rung.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The osteon, in cross section

BRAIN DUMP FIRST

One minute. Every structure in compact bone and how a deep osteocyte gets fed.

Draw one osteon large. Central canal in the middle, concentric lamellae as rings, lacunae between the rings with an osteocyte in each, and canaliculi as fine lines linking them. Add a perforating canal crossing to a neighbouring osteon, and squeeze interstitial lamellae into the gaps between osteons.

In your notebook, head the page **M2–2 The osteon, in cross section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can trace with your finger the route a nutrient takes from the central canal to the outermost osteocyte.

3. Explain why a twisting force breaks a long bone more readily than a force along its length.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The growth plate, zone by zone

BRAIN DUMP FIRST

One minute. The five zones in order and what happens in each.

Draw a strip of growth plate with the epiphysis at the top and the diaphysis at the bottom. Divide it into five bands. Draw the chondrocytes changing as they descend: resting, then stacked in columns, then swollen, then a calcifying matrix, then bone. Label each zone.

In your notebook, head the page **M2–3 The growth plate, zone by zone** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain why a fracture through this plate threatens future growth, using only your drawing.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The three cartilages

GRID

2 Gross anatomy of a long bone

HUB

3 The osteon

HUB

4 Compact against spongy bone

SPLIT

5 The four bone cells

LADDER

6 Intramembranous against endochondral ossification

SPLIT

7 The five zones of the growth plate

LADDER

Pre-work · Axial Skeleton

Axial Skeleton, the skull and the spine. Read both Part A chapters first. This is the largest volume of naming in the course, so the organizing is what makes it survivable.

Module 2 · 2 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the skull and spine chapters into mini visuals. Eight chunks. The grids do the heavy lifting here.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A loose vertebra has a transverse foramen on each transverse process. Name the region and say how you know.

Answer. Cervical.

Reasoning. Go to the giveaway feature first, not the size or the shape. Only cervical vertebrae have transverse foramina, because only they carry the vertebral arteries up to the brain. Thoracic vertebrae are placed by costal facets, and lumbar by their heavy body and blunt spinous process. So one feature settles it, and the other features are only confirmation.

takes longer than three minutes you are copying instead of organizing.

1. **SPLIT**

   Cranial against facial bones

   One split, eight on one branch and fourteen on the other. Mark which are single and which are paired. This is the frame for the whole skull.

2. **GRID**

   The four sutures

   Suture down the side, bones joined and location across the top. Add the pterion as a fifth row and mark why it matters.

3. **GRID**

   The major foramina

   Foramen down the side, bone and what passes through across the top. Nine rows. Star the four you will confuse.

4. **HUB**

   The sphenoid

   Sphenoid at the hub, because it touches every other cranial bone. Spoke to body, sella turcica, greater and lesser wings, pterygoid processes, optic canal, superior orbital fissure.

5. **GRID**

   The four fontanelles

   Fontanelle down the side, location and closing age across the top. Four rows, and the ages are the part people lose.

6. **LADDER**

   The five regions of the spine

   Five rungs, cervical at the top. Write the count beside each rung and the two fused regions at the bottom.

DRAWING 1

The skull, lateral view

BRAIN DUMP FIRST

Two minutes. Every bone you can name on the side of the skull, and the sutures between them.

Draw the skull in profile large enough to label. Outline each bone visible from the side and shade them differently: frontal, parietal, temporal, occipital, sphenoid, zygomatic, maxilla, nasal, lacrimal, mandible. Draw the four major sutures. Mark the pterion where four bones meet, the mastoid and styloid processes, the external acoustic meatus, and the zygomatic arch.

In your notebook, head the page **M2–1 The skull, lateral view** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can put your finger on the pterion and say which four bones meet there and what runs beneath it.

Set A · Name the bone and the opening

1. The cribriform plate belongs to the ______________________ and passes ______________________.
2. The pituitary gland sits in the ______________________ of the ______________________.
3. The foramen magnum passes the ______________________ and the ______________________.
4. The only freely movable bone of the skull is the ______________________.
5. Name the seven bones that build the orbit.

   ____

   ____

Set B · Predict it

1. A blow lands on the pterion. Predict the injury and name the vessel involved.

   ____

   ____

2. An infant has a bulging anterior fontanelle. Say what it may signal, and what a sunken one may signal instead.

   ____

   ____

3. A tooth infection in the upper jaw spreads. Name the cavity most at risk and say why.

   ____

   ____

4. A patient has a herniated disc at L4 to L5 with pain travelling down the leg. Name the two parts of the disc and explain the path of the pain.

   ____

   ____

Set C · Sort and classify

1. Sort as true, false, or floating: ribs 3, 9, 11.

   ____

7 **GRID**

Cervical, thoracic, lumbar vertebrae

Region down the side. Body shape, spinous process, and the one giveaway feature across the top. Three rows. The giveaway column is what lets you place a loose vertebra.

8 **CHAIN**

The four curvatures, in developmental order

Thoracic and sacral present at birth, then cervical when the head lifts, then lumbar when the child walks. Arrows carry the milestones.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

A typical vertebra, from above

BRAIN DUMP FIRST

One minute. Every part of a vertebra, and which region it belongs to.

Draw one vertebra looking down on it. Body at the front, then the pedicles, laminae, and the arch closing behind, with the vertebral foramen in the middle. Add the spinous process pointing back and the transverse processes to the sides, then the superior articular processes. Now redraw it twice more in a second colour, once as cervical with transverse foramina and once as thoracic with costal facets.

In your notebook, head the page **M2–2 A typical vertebra, from above** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

Someone hands you a loose vertebra and you can name its region from one feature without hesitating.

2. Name the four curvatures and mark each primary or secondary, with the milestone that produces the secondary ones.

3. Compare the atlanto-occipital and atlanto-axial joints by the movement each produces.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The thoracic cage

BRAIN DUMP FIRST

One minute. The ribs, how they attach, and the parts of the sternum.

Draw the cage from the front. Sternum in three parts with the jugular notch, sternal angle, and xiphoid process labelled. Draw the twelve rib pairs and their costal cartilages, then bracket them: true, false, and floating. Mark the sternal angle at the level of the second rib.

In your notebook, head the page **M2–3 The thoracic cage** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can count ribs from the sternal angle the way a clinician does, and say where you would stop for each group.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Cranial against facial bones SPLIT

2 The four sutures GRID

3 The major foramina GRID

4 The sphenoid HUB

5 The four fontanelles
GRID

6 The five regions of the spine
LADDER

7 Cervical, thoracic, lumbar vertebrae
GRID

8 The four curvatures, in developmental order
CHAIN

Pre-work · Appendicular Skeleton

Appendicular Skeleton, the upper and lower limbs. Read both Part A chapters first. Learn the markings vocabulary before the markings, because it repeats on every bone.

Module 2 · 3 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the two limb chapters into mini visuals. Eight chunks, and the two comparison splits are worth the most.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

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PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient falls on an outstretched hand and has tenderness in the anatomical snuffbox. Name the likely fracture and say why it matters.

Answer. A scaphoid fracture.

Reasoning. Start from the location, then ask what lies under it. The snuffbox is the hollow at the base of the thumb, and the scaphoid sits in its floor, so tenderness there points at that bone. It matters because the scaphoid has a poor blood supply, so a fracture missed on a first X-ray can fail to heal. That is a structural fact producing a clinical rule: suspect it, splint it, image it again.

takes longer than three minutes you are copying instead of organizing.

1 **GRID**

Bone markings vocabulary

Marking down the side, type and example across the top. Nine rows. Sort each as a projection, a depression, or an opening. Do this one first.

2 **CHAIN**

Upper limb, proximal to distal

Girdle, arm, forearm, hand, with the bones under each link. Then run a second chain underneath for the lower limb and line the two up so the parallels show.

3 **HUB**

The scapula

Scapula at the hub. Spoke to the three borders, the spine, acromion, coracoid process, glenoid cavity, and the three fossae.

4 **SPLIT**

Radius against ulna

One split. Side of the forearm, proximal markings, distal markings on each branch. This is the pair most people lose.

5 **HUB**

The humerus

Humerus down the middle. Proximal markings on one side, distal on the other. Mark which distal surface meets which forearm bone.

6 **HUB**

The coxal bone

Acetabulum at the hub, because all three bones meet there. Spoke out to ilium, ischium, and pubis, then to the markings on each.

DRAWING 1

The humerus, both views

BRAIN DUMP FIRST

Two minutes. Every marking on the humerus, proximal and distal.

Draw the humerus twice, anterior and posterior. On the anterior view mark head, anatomical and surgical necks, greater and lesser tubercles, intertubercular sulcus, deltoid tuberosity, capitulum, trochlea, and both epicondyles. On the posterior view add the olecranon fossa and the radial groove, and mark where the radial nerve runs.

In your notebook, head the page **M2–1 The humerus, both views** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which forearm bone meets the capitulum and which meets the trochlea without pausing.

Set A · Name the marking

1. The only bony attachment of the upper limb to the axial skeleton is the ______________________.
2. The socket that receives the head of the humerus is the ______________________.
3. The point of the elbow is the ______________________ of the ______________________.
4. The deep cup where the three hip bones meet is the ______________________.
5. The tarsal that receives the body weight from the tibia is the ______________________.

Set B · Predict it

1. A fracture at the surgical neck of the humerus. Name the nerve at risk and say why it is checked.

____

____

2. A patient cannot feel the little finger after knocking the medial elbow. Name the nerve and where it runs.

____

____

3. An older adult falls and cannot bear weight on one leg, with the limb shortened and rotated. Name the likely fracture site and the structure whose blood supply is threatened.

____

____

4. Surgeons harvest bone from the fibula rather than the tibia. Explain why, using weight bearing.

____

____

Set C · Tell it apart

1. Give one feature that tells the radius from the ulna at the elbow, and one at the wrist.

____

____

7 SPLIT

Tibia against fibula

One split. Side, whether it bears weight, and the malleolus each one forms. Three branches, and the weight-bearing branch explains the rest.

8 GRID

Carpals and tarsals

Two grids side by side. Carpals in two rows of four with the mnemonic, tarsals in their arrangement with the three C's around the N and T.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The forearm in supination and pronation

BRAIN DUMP FIRST

One minute. Radius and ulna, which is which, and what moves.

Draw the two bones side by side in supination, palm forward, and label radius lateral and ulna medial. Then draw them again in pronation with the radius crossed over the ulna. Mark the proximal and distal radioulnar joints and the interosseous membrane in both.

In your notebook, head the page **M2–2 The forearm in supination and pronation** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain which bone moves and which stays put, and name the joints that allow it.

2. Name the carpals in both rows in order, using the mnemonic, and say which is fractured most often.

3. Compare the pectoral and pelvic girdles by stability and range of motion, and say what causes the difference.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The coxal bone and the knee

BRAIN DUMP FIRST

One minute. The three fused bones, then the ligaments of the knee.

Draw the coxal bone from the side, shading ilium, ischium, and pubis differently, and mark the acetabulum where all three meet plus the obturator foramen. Then draw the knee from the front with the patella removed: femoral condyles, tibial plateau, both menisci, both cruciate ligaments crossing, and both collateral ligaments.

In your notebook, head the page **M2–3 The coxal bone and the knee** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name the three structures of the unhappy triad on your own knee drawing and say what force causes it.

Mini visual sheet

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1 Bone markings vocabulary

GRID

2 Upper limb, proximal to distal

CHAIN

3 The scapula

HUB

4 Radius against ulna

SPLIT

5 The humerus

HUB

6 The coxal bone

HUB

7 Tibia against fibula

SPLIT

8 Carpals and tarsals

GRID

Pre-work · Joints

Articulations and Joints. Read the Part A chapter first. Every tagged joint can be asked four ways, so practise answering all four every time.

Module 2 · 4 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
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3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the joints chapter into mini visuals. Seven chunks. The two classification grids are the spine of the whole chapter.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

Classify the pubic symphysis by structure and by function, and name the tissue involved.

Answer. Structurally a cartilaginous joint, specifically a symphysis. Functionally an amphiarthrosis. The tissue is fibrocartilage.

Reasoning. Run the two classifications separately, then check they agree. Structure first: is there a synovial cavity? No, so it is fibrous or cartilaginous. What holds the bones? A broad flat disc of fibrocartilage, which is the definition of a symphysis. Function second: fibrocartilage allows a little give, so slightly movable, an amphiarthrosis. The two answers are consistent, which is the check that you classified it correctly.

takes longer than three minutes you are copying instead of organizing.

1. **GRID**

   Structure crossed with function

   Fibrous, cartilaginous, synovial down the side. Cavity present, tissue holding the bones, and usual functional class across the top. This one grid answers most exam questions in this chapter.

2. **HUB**

   Fibrous joints

   Fibrous at the hub. Spoke to suture, syndesmosis, gomphosis, interosseous membrane, synostosis. One example on each spoke.

3. **SPLIT**

   Synchondrosis against symphysis

   One split. Cartilage type, mobility, and two examples on each branch. The cartilage type is what decides everything else.

4. **HUB**

   A synovial joint in section

   Joint cavity at the hub. Spoke to articular cartilage, capsule with its two membranes, ligaments, synovial fluid, fat pads, menisci, labrum, bursae.

5. **GRID**

   The six synovial types

   Type down the side. Surface shape, axes, and one example across the top. Six rows, and the axes column is what people lose.

6. **GRID**

   The movements

   Two grids. Angular movements in one, special movements in the other. Draw a tiny arrow showing each direction rather than writing a definition.

DRAWING 1

A synovial joint, sectioned

BRAIN DUMP FIRST

Two minutes. Every structure inside a synovial joint and what each one does.

Draw two bone ends meeting, with a gap between them. Cap each with articular cartilage. Draw the capsule wrapping the joint, with the fibrous membrane outside and the synovial membrane inside, and fill the cavity with synovial fluid. Add a meniscus between the surfaces, a fat pad, and a bursa outside the capsule. Label all of it.

In your notebook, head the page **M2–1 A synovial joint, sectioned** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name the four jobs of synovial fluid while pointing at where it sits.

Set A · Classify it

1. A tooth in its socket. Structural class ______________ Functional class ______________
2. The elbow. Synovial type ______________ Axes ______________
3. The epiphyseal plate. Structural class ______________ Functional class ______________
4. The carpometacarpal joint of the thumb. Synovial type ______________ Axes ______________
5. The atlantoaxial joint. Synovial type ______________ Movement it produces ______________

Set B · Name the movement

1. Turning the sole of the foot inward is ______________________.
2. Moving the thumb across the palm to touch the fingertips is ______________________.
3. Turning the palm to face posteriorly is ______________________.
4. Bending the head sideways toward the shoulder is ______________________.
5. Name the movements possible at a ball-and-socket joint.

   ____

   ____

Set C · Predict it

1. A lateral blow lands on a planted knee. Name the three structures likely torn and the name of the injury.

   ____

   ____

2. A patient has an unstable shoulder after an injury. Explain why the ligaments alone do not stabilize it, and name what does.

   ____

   ____

7 SPLIT

Osteoarthritis against rheumatoid arthritis

One split. Cause, which joints, and whether it is symmetrical. Three branches each. This pair appears on every exam.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The knee, anterior and from above

BRAIN DUMP FIRST

Two minutes. Every ligament and both menisci.

Draw the knee from the front with the patella lifted away: femoral condyles, tibial plateau, both collateral ligaments outside, both cruciates crossing inside. Then draw the tibial plateau from above and put the C-shaped medial meniscus and the O-shaped lateral meniscus in place. Mark the three structures of the unhappy triad.

In your notebook, head the page **M2–2 The knee, anterior and from above** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain what stops the tibia sliding forward, and what happens when it fails.

3. A knee swells immediately after an injury, and another swells the next day. Say what each timing suggests.

4. Distinguish osteoarthritis from rheumatoid arthritis by cause, joints affected, and symmetry.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The shoulder, and why it dislocates

BRAIN DUMP FIRST

One minute. The glenohumeral joint and what actually holds it together.

Draw the humeral head against the glenoid cavity and mark how much larger the head is than the socket. Add the glenoid labrum deepening the rim, the loose capsule, and the coracohumeral and glenohumeral ligaments. Then draw the four rotator cuff muscles gripping the head, and label them.

In your notebook, head the page **M2–3 The shoulder, and why it dislocates** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say why the ligaments are not what stabilizes this joint, and name the four muscles that are.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Structure crossed with function
GRID

2 Fibrous joints
HUB

3 Synchondrosis against symphysis
SPLIT

4 A synovial joint in section
HUB

5 **The six synovial types** GRID

6 **The movements** GRID

7 **Osteoarthritis against rheumatoid arthritis** SPLIT

PART C

Lab Structure List.

Part C of the Module 2 packet. These are the lab structures and concepts to learn for the lab portion of Exam 1. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

Bone Tissue and the Long Bone

MICROSCOPIC AND GROSS

Osseous tissue on the model and the slide, then the whole long bone. Two questions on this sheet ask you to name a cartilage type, so answer them from the tissue chapter.

☐ Osseous tissue, on the bone tissue model

STRUCTURES TO IDENTIFY

- ☐ Osteon
- ☐ Central (Haversian) canal
- ☐ Volkmann's canal
- ☐ Canaliculi
- ☐ Lacunae
- ☐ Osteocytes
- ☐ Inner circumferential lamellae
- ☐ Outer circumferential lamellae
- ☐ Interstitial lamellae
- ☐ Concentric lamellae
- ☐ Medullary cavity
- ☐ Periosteum, inner osteogenic layer
- ☐ Periosteum, outer fibrous layer
- ☐ Sharpey's fibers

☐ Gross anatomy of a long bone

STRUCTURES TO IDENTIFY

- ☐ Epiphysis
- ☐ Metaphysis
- ☐ Diaphysis
- ☐ Medullary cavity
- ☐ Endosteum
- ☐ Periosteum, osteogenic layer
- ☐ Periosteum, fibrous layer
- ☐ Articular cartilage
- ☐ Epiphyseal line, in adults
- ☐ Epiphyseal plate, in children
- ☐ Compact bone
- ☐ Spongy bone
- ☐ Collagen fibers between the lamellae
- ☐ Red marrow
- ☐ Yellow marrow

Articular cartilage. Which type of cartilage is it? ____

Epiphyseal plate. Which type of cartilage is it? ____

CHAPTER 2

Axial Skeleton: The Skull

CRANIAL AND FACIAL BONES, SUTURES, FORAMINA

The largest single block of identifications in the course. Work it view by view, not bone by bone, because that is how it is tagged.

☐ Cranial and facial bones

STRUCTURES TO IDENTIFY

- ☐ Frontal
- ☐ Occipital
- ☐ Parietal
- ☐ Temporal
- ☐ Sphenoid
- ☐ Ethmoid
- ☐ Maxilla
- ☐ Zygomatic
- ☐ Nasal
- ☐ Lacrimal
- ☐ Mandible

☐ Sutures

STRUCTURES TO IDENTIFY

- ☐ Coronal
- ☐ Sagittal
- ☐ Squamosal
- ☐ Lambdoid

☐ Lateral skull

STRUCTURES TO IDENTIFY

- ☐ Glabella
- ☐ Supraorbital foramen or notch
- ☐ Supraorbital margin
- ☐ Zygomatic arch
- ☐ Zygomatic process of the temporal bone
- ☐ Temporal process of the zygomatic bone
- ☐ Mastoid process
- ☐ External acoustic (auditory) meatus
- ☐ Styloid process
- ☐ Articular tubercle
- ☐ Alveolar process
- ☐ Anterior nasal spine
- ☐ Infraorbital foramen

☐ Anterior skull

STRUCTURES TO IDENTIFY

- ☐ Supraorbital foramen or notch
- ☐ Supraorbital margin
- ☐ Glabella
- ☐ Perpendicular plate of the ethmoid
- ☐ Vomer
- ☐ Middle nasal concha
- ☐ Inferior nasal concha
- ☐ Middle nasal meatus
- ☐ Inferior nasal meatus
- ☐ Anterior nasal spine
- ☐ Alveolar process
- ☐ Infraorbital foramen
- ☐ Nasion

☐ Posterior skull

STRUCTURES TO IDENTIFY

- ☐ Superior nuchal line
- ☐ Inferior nuchal line
- ☐ External occipital protuberance
- ☐ Vomer
- ☐ Occipital condyles
- ☐ Palatine bone
- ☐ Palatine process of the maxilla
- ☐ Incisive fossa and foramen
- ☐ Mastoid process

☐ Base of the skull, exterior

STRUCTURES TO IDENTIFY

- ☐ Incisive foramen
- ☐ Transverse palatine suture
- ☐ Median palatine suture
- ☐ Palatine bone
- ☐ Vomer
- ☐ Medial pterygoid plate
- ☐ Lateral pterygoid plate
- ☐ Foramen ovale
- ☐ Foramen spinosum
- ☐ Foramen lacerum
- ☐ Foramen rotundum
- ☐ Carotid canal
- ☐ Jugular foramen
- ☐ Hypoglossal canal
- ☐ Foramen magnum
- ☐ Inferior nuchal line
- ☐ Superior nuchal line
- ☐ External occipital protuberance
- ☐ Mastoid process
- ☐ Articular tubercle
- ☐ Occipital condyles
- ☐ Styloid process
- ☐ Mandibular fossa
- ☐ Zygomatic arch
- ☐ Inferior orbital fissure
- ☐ Choana

☐ Base of the skull, interior

STRUCTURES TO IDENTIFY

- ☐ Anterior cranial fossa
- ☐ Middle cranial fossa
- ☐ Posterior cranial fossa
- ☐ Frontal crest
- ☐ Frontal sinus
- ☐ Cribriform plate of the ethmoid
- ☐ Crista galli
- ☐ Olfactory foramina
- ☐ Lesser wing of the sphenoid
- ☐ Greater wing of the sphenoid
- ☐ Hypophyseal fossa (sella turcica)
- ☐ Dorsum sellae
- ☐ Tuberculum sellae
- ☐ Internal acoustic (auditory) meatus
- ☐ Jugular foramen
- ☐ Internal occipital protuberance
- ☐ Hypoglossal canal
- ☐ Foramen lacerum
- ☐ Foramen magnum
- ☐ Foramen spinosum
- ☐ Foramen ovale
- ☐ Optic canal
- ☐ Anterior clinoid processes

Which structure sits in the hypophyseal fossa? ______________________________

☐ Ethmoid bone

STRUCTURES TO IDENTIFY

- ☐ Crista galli
- ☐ Perpendicular plate
- ☐ Cribriform plate
- ☐ Olfactory foramina

☐ Sphenoid bone

STRUCTURES TO IDENTIFY

- ☐ Medial pterygoid plate
- ☐ Lateral pterygoid plate
- ☐ Hypophyseal fossa (sella turcica)
- ☐ Dorsum sellae
- ☐ Tuberculum sellae
- ☐ Greater wing
- ☐ Lesser wing
- ☐ Superior orbital fissure

☐ Orbit

STRUCTURES TO IDENTIFY

- ☐ Sphenoid bone
- ☐ Lacrimal bone
- ☐ Lacrimal canal
- ☐ Optic canal (foramen)
- ☐ Superior orbital fissure
- ☐ Inferior orbital fissure

☐ Mandible

STRUCTURES TO IDENTIFY

- ☐ Condylar process (condyle)
- ☐ Coronoid process
- ☐ Mandibular notch
- ☐ Mental protuberance
- ☐ Mental foramen
- ☐ Alveolar process
- ☐ Alveolar margin
- ☐ Body
- ☐ Ramus
- ☐ Mandibular foramen
- ☐ Angle of the mandible

☐ Hyoid bone

STRUCTURES TO IDENTIFY

- ☐ Body
- ☐ Greater cornu (greater horn)
- ☐ Lesser cornu (lesser horn)

☐ Nasal septum

STRUCTURES TO IDENTIFY

- ☐ Septal cartilage
- ☐ Perpendicular plate of the ethmoid
- ☐ Vomer

☐ Fetal skull

STRUCTURES TO IDENTIFY

- ☐ Anterior fontanel
- ☐ Posterior fontanel
- ☐ Right and left anterolateral fontanel
- ☐ Right and left posterolateral fontanel

CHAPTER 3

Axial Skeleton: Spine and Thoracic Cage

VERTEBRAE, SACRUM, RIBS, STERNUM

Identify the segment, then the region, then the individual vertebra. The regional features are what let you place a single loose vertebra.

☐ Spinal segments

STRUCTURES TO IDENTIFY

- ☐ Cervical spine, C1 to C7
- ☐ Thoracic spine, T1 to T12
- ☐ Lumbar spine, L1 to L5
- ☐ Sacrum, S1 to S5 fused
- ☐ Coccyx, fused

☐ Normal spinal curves

Label each as primary or secondary. Know which are present at birth, and which developmental milestone produces each secondary curve, such as holding the head up, sitting, and walking.

STRUCTURES TO IDENTIFY

- ☐ Cervical
- ☐ Thoracic
- ☐ Lumbar
- ☐ Sacrococcygeal

☐ Abnormal spinal curves

Define each as well as identify it.

STRUCTURES TO IDENTIFY

- ☐ Scoliosis
- ☐ Lordosis
- ☐ Kyphosis

☐ C1, the atlas

STRUCTURES TO IDENTIFY

- ☐ Anterior tubercle
- ☐ Posterior tubercle
- ☐ Facet for the dens
- ☐ Anterior arch
- ☐ Posterior arch
- ☐ Superior articular facet and process
- ☐ Inferior articular facet and process
- ☐ Transverse process
- ☐ Transverse foramen
- ☐ Vertebral foramen

☐ C2, the axis

STRUCTURES TO IDENTIFY

- ☐ Dens
- ☐ Superior articular facet and process
- ☐ Inferior articular facet and process
- ☐ Body
- ☐ Transverse process
- ☐ Transverse foramen
- ☐ Vertebral foramen
- ☐ Spinous process
- ☐ Vertebral arch

☐ Typical cervical vertebra

STRUCTURES TO IDENTIFY

- ☐ Vertebral foramen
- ☐ Transverse process
- ☐ Transverse foramen
- ☐ Body
- ☐ Pedicle
- ☐ Lamina
- ☐ Spinous process
- ☐ Vertebral arch
- ☐ Superior articular facet and process
- ☐ Inferior articular facet and process

☐ Typical thoracic and lumbar vertebra

STRUCTURES TO IDENTIFY

- ☐ Body
- ☐ Superior articular facet and process
- ☐ Inferior articular facet and process
- ☐ Costal facet, thoracic only
- ☐ Transverse process
- ☐ Spinous process
- ☐ Lamina
- ☐ Pedicle
- ☐ Vertebral arch
- ☐ Vertebral foramen

☐ Placing a vertebra on the articulated column

Identify specific vertebrae on an articulated column, for example C7, T5, L1. Then work the other way: from the features alone, name the region. A transverse foramen means cervical. A costal facet means thoracic. A bifid spinous process means cervical. A spinous process pointing straight back means lumbar.

☐ Rib

STRUCTURES TO IDENTIFY

- ☐ Head
- ☐ Neck
- ☐ Shaft

☐ Sacrum and coccyx

STRUCTURES TO IDENTIFY

- ☐ Sacral promontory
- ☐ Superior articular process and facet
- ☐ Anterior sacral foramina
- ☐ Coccyx
- ☐ Posterior sacral foramina
- ☐ Sacral hiatus
- ☐ Median sacral crest
- ☐ Lateral sacral crest
- ☐ Articular surface
- ☐ Base of the sacrum
- ☐ Apex of the sacrum
- ☐ Sacral tuberosity

☐ Sternum

STRUCTURES TO IDENTIFY

- ☐ Manubrium
- ☐ Sternal angle
- ☐ Body
- ☐ Xiphoid process
- ☐ Jugular notch
- ☐ Clavicular notch
- ☐ Costal notch

☐ Intervertebral disc

STRUCTURES TO IDENTIFY

- ☐ Annulus fibrosus
- ☐ Nucleus pulposus

CHAPTER 4

Appendicular Skeleton: Pectoral Girdle and Upper Limb

CLAVICLE TO PHALANGES

Girdle first, then limb, then hand. The carpals have a memory tool built into the list.

☐ Clavicle

STRUCTURES TO IDENTIFY

- ☐ Sternal end
- ☐ Acromial end
- ☐ Conoid tubercle

☐ Scapula

STRUCTURES TO IDENTIFY

- ☐ Acromion
- ☐ Coracoid process
- ☐ Spine of the scapula
- ☐ Infraspinous fossa
- ☐ Supraspinous fossa
- ☐ Subscapular fossa
- ☐ Glenoid cavity
- ☐ Supraglenoid tubercle
- ☐ Infraglenoid tubercle
- ☐ Superior border
- ☐ Superior angle
- ☐ Suprascapular notch
- ☐ Axillary (lateral) border
- ☐ Vertebral (medial) border
- ☐ Inferior angle

☐ Humerus

STRUCTURES TO IDENTIFY

- ☐ Head
- ☐ Anatomical neck
- ☐ Surgical neck
- ☐ Greater tubercle
- ☐ Lesser tubercle
- ☐ Intertubercular sulcus (bicipital groove)
- ☐ Deltoid tuberosity
- ☐ Radial groove
- ☐ Lateral epicondyle
- ☐ Lateral supracondylar ridge
- ☐ Medial epicondyle
- ☐ Medial supracondylar ridge
- ☐ Trochlea
- ☐ Coronoid fossa
- ☐ Capitulum
- ☐ Radial fossa
- ☐ Olecranon fossa

☐ Radius

STRUCTURES TO IDENTIFY

- ☐ Head
- ☐ Neck
- ☐ Styloid process
- ☐ Radial tuberosity
- ☐ Ulnar notch

☐ Ulna

STRUCTURES TO IDENTIFY

- ☐ Head
- ☐ Styloid process
- ☐ Ulnar tuberosity
- ☐ Radial notch
- ☐ Coronoid process
- ☐ Trochlear notch
- ☐ Olecranon process

☐ Carpals, on an articulated wrist

Memory tool: So Long To Pinky, proximal row lateral to medial. Here Comes The Thumb, distal row medial to lateral.

STRUCTURES TO IDENTIFY

- ☐ Scaphoid
- ☐ Lunate
- ☐ Triquetrum (triangular)
- ☐ Pisiform
- ☐ Hamate
- ☐ Capitate
- ☐ Trapezoid
- ☐ Trapezium

☐ Metacarpals and phalanges

The first digit has no middle phalanx.

STRUCTURES TO IDENTIFY

- ☐ Metacarpals 1 to 5, lateral to medial
- ☐ Proximal phalanges
- ☐ Middle phalanges
- ☐ Distal phalanges

CHAPTER 5

Appendicular Skeleton: Pelvic Girdle and Lower Limb

COXAL BONE TO PHALANGES

The coxal bone is three fused bones, so learn it as three regions that meet at the acetabulum.

☐ Coxal bone, the ilium region

STRUCTURES TO IDENTIFY

- ☐ Iliac crest
- ☐ Auricular surface
- ☐ Iliac fossa
- ☐ Greater sciatic notch
- ☐ Lesser sciatic notch
- ☐ Ala
- ☐ Anterior superior iliac spine (ASIS)
- ☐ Anterior inferior iliac spine (AIIS)
- ☐ Posterior superior iliac spine (PSIS)
- ☐ Posterior inferior iliac spine (PIIS)

☐ Ischium

STRUCTURES TO IDENTIFY

- ☐ Ischial spine
- ☐ Ischial tuberosity
- ☐ Lesser sciatic notch
- ☐ Ramus of the ischium

☐ Pubis

STRUCTURES TO IDENTIFY

- ☐ Pubic body
- ☐ Superior ramus
- ☐ Inferior ramus
- ☐ Pubic crest
- ☐ Pubic tubercle

☐ Landmarks and joints of the pelvis

STRUCTURES TO IDENTIFY

- ☐ Acetabulum
- ☐ Obturator foramen
- ☐ Sacroiliac joint
- ☐ Pubic symphysis

☐ Femur

STRUCTURES TO IDENTIFY

- ☐ Head
- ☐ Fovea capitis
- ☐ Neck
- ☐ Greater trochanter
- ☐ Lesser trochanter
- ☐ Intertrochanteric line
- ☐ Intertrochanteric crest
- ☐ Gluteal tuberosity
- ☐ Linea aspera, lateral lip
- ☐ Linea aspera, medial lip
- ☐ Lateral supracondylar line
- ☐ Medial supracondylar line
- ☐ Adductor tubercle
- ☐ Lateral epicondyle
- ☐ Medial epicondyle
- ☐ Lateral condyle
- ☐ Medial condyle
- ☐ Patellar surface
- ☐ Intercondylar fossa

☐ Patella

☐ Tibia

STRUCTURES TO IDENTIFY

- ☐ Lateral condyle
- ☐ Medial condyle
- ☐ Intercondylar eminence
- ☐ Tibial tuberosity
- ☐ Fibular notch
- ☐ Anterior crest (border)
- ☐ Medial malleolus

☐ Fibula

STRUCTURES TO IDENTIFY

- ☐ Head of the fibula
- ☐ Lateral malleolus

☐ Tarsals

Memory device: the three C's surround the N and T.

STRUCTURES TO IDENTIFY

- ☐ Talus
- ☐ Calcaneus
- ☐ Navicular
- ☐ Medial cuneiform
- ☐ Intermediate cuneiform
- ☐ Lateral cuneiform
- ☐ Cuboid

☐ Metatarsals and phalanges

The first digit, the hallux, has only a proximal and a distal phalanx. Toes 2 to 5 have all three.

STRUCTURES TO IDENTIFY

- ☐ Metatarsals I to V, medial to lateral
- ☐ Proximal phalanges
- ☐ Middle phalanges
- ☐ Distal phalanges

CHAPTER 6

Joints

CLASSIFICATION, NAMED JOINTS, THE KNEE

For any tagged joint you may be asked four things. Practise answering all four every time, not just the name.

For any joint you are given, be ready to

1. Classify it as fibrous, cartilaginous, or synovial.
2. Name the type of synovial joint, for example hinge, ball and socket, saddle.
3. Classify it by movement: synarthrosis (immovable), amphiarthrosis (slightly movable), diarthrosis (freely movable).
4. Name the specific movements that occur there, if any. Flexion, extension, internal rotation, and so on.

☐ Named joints, know these when tagged

STRUCTURES TO IDENTIFY

- ☐ Atlantoaxial
- ☐ Atlanto-occipital
- ☐ Glenohumeral
- ☐ Acromioclavicular
- ☐ Sternoclavicular
- ☐ Manubriosternal
- ☐ Radioulnar
- ☐ Talocrural
- ☐ Tibiofibular
- ☐ Temporomandibular
- ☐ Sacroiliac
- ☐ Pubic symphysis

☐ A typical synovial joint

STRUCTURES TO IDENTIFY

- ☐ Articular cartilage
- ☐ Synovial membrane
- ☐ Joint cavity
- ☐ Synovial fluid

☐ The knee joint

STRUCTURES TO IDENTIFY

- ☐ Femur
- ☐ Tibia
- ☐ Fibula
- ☐ Patella
- ☐ Patellar ligament
- ☐ Quadriceps tendon
- ☐ Medial (tibial) collateral ligament
- ☐ Lateral (fibular) collateral ligament
- ☐ Medial meniscus
- ☐ Lateral meniscus
- ☐ Anterior cruciate ligament
- ☐ Posterior cruciate ligament